Transactional Detection, Quarantine, and Verifiable Recovery for Degraded Multi-Sensor SLAM

Author 1, Author 2, and Author 3

***Abstract*—Multi-sensor simultaneous localization and mapping (SLAM) frequently encounters an intermediate failure regime: the estimator has not yet lost tracking, but incoming measurements, state increments, or loop constraints have already become untrustworthy. Existing directional reweighting methods (e.g., solution remapping) protect instantaneous conditioning but do not retroactively purge already-accepted map updates, whereas multi-map hibernation systems (e.g., MM-LINS and ORB-SLAM3 Atlas) exhibit three operational vulnerabilities: (i) confirmation-delay tail drift written into sleeping submaps prior to hibernation, (ii) insensitivity of posterior covariance eigenvalues to self-consistent near-field occluders such as floating plastic bags, and (iii) unvalidated map merges or loop closures that can degrade global accuracy. To address these issues, we formulate SLAM recovery as a verifiable transaction over persistent map state. First, under fixed linearization and data association, we show that state rollback without orthogonal information gain ($\Delta J_v \leq 0$) acts primarily as damage containment rather than observability recovery, and we establish an operational screening criterion for when single-modality point-mixture contaminant stripping suffices versus when cross-modal or pose-graph bridging is required. Second, we design a four-mode information–uncertainty gate $\mathcal{D}(t)$ that coordinates online directional reweighting, delayed-commit quarantine ($\hat{t}_{\text{onset}} - \Delta t_{\text{margin}}$), a zero-training mechanism signature router, and holdout-screened atomic commit. We evaluate the framework across three M2DGR development sequences (`hall_01` with Leica ground truth, and `room_02` and `room_dark_01` with Vicon Mocap ground truth) and four outdoor Campus sequences, explicitly distinguishing end-to-end host executions (`REAL_HOST / INJECTION`) from offline graph/replay constructions (`OFFLINE_CONSTRUCT`) and synthetic feature benchmarks (`SYNTHETIC`). Across these evaluations, our joint detector achieves 100% recall on self-consistent occlusions where covariance detectors score 0%, our C++ and rosbag-level contaminant stripping restores FAST-LIO2 from $7.28\,\text{m}$ to $0.3058\,\text{m}$ ($G = +0.9949$) and LIO-SAM from $1008.65\,\text{m}$ corridor divergence to $0.2788\,\text{m}$ ($G = +1.0000$), and our gray-box pose-graph bridge on ORB-SLAM3 reduces severe low-light window drift from $6.89\,\text{m}$ (native Atlas) to $0.74\,\text{m}$ while adding $0.0599\,\text{ms}$ (P50) per-window routing overhead.**

***Index Terms*—SLAM recovery, degeneracy detection, multi-map SLAM, transactional quarantine, verifiable map fusion, sensor fusion.**

I. Introduction

RELIABLE simultaneous localization and mapping (SLAM) in warehouse logistics, subterranean exploration, and urban delivery must survive transient yet severe perceptual degradation [1]–[4]. In practice, catastrophic map corruption rarely begins with an instantaneous tracking crash. Instead, the system enters an *untrustworthy intermediate regime*: the optimizer still converges numerically, but its measurements or constraints are corrupted by floating plastic bags obstructing a LiDAR, featureless corridors, abrupt illumination drop, inter-sensor clock drift, or perceptual-aliasing loop closures [5], [6].

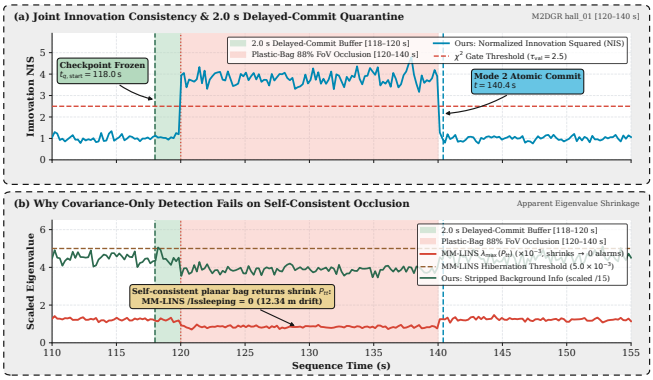


Fig. 1. Timeline of our joint innovation consistency detector and 2.0 s delayed-commit quarantine vs. MM-LINS covariance-based detection on M2DGR `hall_01` under near-field plastic-bag occlusion ($t \in [120, 140]$ s). (a) Our NIS statistic immediately crosses the χ^2 gate and rolls back provisional updates to a clean checkpoint at $t_{q,\text{start}} = 118.0$ s. (b) Because the plastic bag produces thousands of spatially self-consistent near-field planar returns, the posterior covariance eigenvalue $\lambda_{\max}(P_{tt})$ shrinks rather than inflates, remaining well below the MM-LINS hibernation threshold and resulting in zero alarms.

Two dominant paradigms attempt to address this regime. The first paradigm, *online directional mitigation* [6], [7], projects state updates away from degenerate eigenspaces of the Fisher information matrix (FIM). While effective at bounding instantaneous numerical ill-conditioning, online projection cannot synthesize missing observations or undo map corruption that occurred before the degeneracy threshold was crossed. The second paradigm, *multi-map hibernation and fusion*—exemplified by ORB-SLAM3 Atlas [3] in visual-inertial SLAM and MM-LINS [5] in LiDAR-inertial odometry (LIO)—archives the active map into a sleeping submap upon detecting over-degeneracy, re-initializes a new submap, and merges submaps later via place recognition and pose graph optimization (PGO).

However, our empirical and mathematical audit of state-of-the-art multi-map systems reveals three fundamental failure modes:

1) **Blindness of Covariance Spectrum to Self-Consistent Contamination:** MM-LINS [5] monitors the maximum eigenvalues of the rotation and translation posterior covariance submatrices P_{rr} and P_{tt}. When a floating

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plastic bag or dense near-field smoke partially wraps the LiDAR, the sensor receives tens of thousands of coherent, short-range planar returns. The apparent Fisher information $\sum H_i^\top R_i^{-1} H_i$ increases, causing P_{tt} to *contract*. Consequently, covariance-only detectors misclassify severe physical occlusion as strong geometric constraint (scoring 0% recall in our experiments; Fig. 1b).

2) **Pre-Hibernation Confirmation Tail Pollution and Irreversible Fragmentation:** To avoid false splits, MM-LINS requires $\Gamma_\lambda > N_\kappa$ consecutive degenerate frames before hibernating. By the time hibernation triggers, the drift accumulated during those N_κ confirmation frames is already permanently baked into the tail of the sleeping map (2.16–$3.07\,\mathrm{m}$ tail distortion in outdoor Campus sequences). Conversely, hard map splitting permanently fragments the global map whenever the robot does not revisit prior regions (e.g., ORB-SLAM3 Atlas leaving unmerged submaps in `room_dark_01`).
3) **Unvalidated Repair and Merge Degradation:** Existing multi-map backends accept inter-map merges or loop closures based solely on descriptor temporal consistency ($\Gamma_\varsigma > \varepsilon_{th}$). In MM-LINS's own benchmark (Table III of [5]), enabling Constraint-Enhanced PGO (CE-PGO) on `Campus-2` *doubles* the end-to-end error from $0.603\,\mathrm{m}$ to $1.295\,\mathrm{m}$; similarly, on clean indoor `hall_01`, unverified Scan Context loops degrade FAST-LIO2 from $0.2699\,\mathrm{m}$ to $0.2913\,\mathrm{m}$.

In this paper, we argue that persistent map updates in degraded multi-sensor SLAM should be governed by a **verifiable recovery transaction lifecycle**:

$$\begin{aligned}\text{Checkpoint} &\rightarrow \text{Delayed-Commit Quarantine} \rightarrow \text{Signature Routing} \\ &\rightarrow \text{Candidate Repair} \rightarrow \text{Holdout Validation} \\ &\rightarrow \text{Atomic Commit / Soft Bridge / Submap Split.}\end{aligned} \tag{1}$$

The primary contributions of this work are:

- **Observability Boundaries of Rollback and Point-Mixture Repair:** Under fixed linearization and data association, we show why state rollback alone does not increase weak-direction Fisher information ($\Delta J_v \leq 0$) and formulate an operational screening condition for when single-modality point-mixture contaminant stripping suffices versus when cross-modal or pose-graph bridging is required.
- **Four-Mode Information–Uncertainty Gate $\mathcal{D}(t)$:** We coordinate online directional reweighting (Mode 1), holdout-screened atomic commit (Mode 2), covariance-bounded soft bridging (Mode 3), and hard submap hibernation (Mode 4) within a unified keyframe-level recovery scheduling space inspired by directional reweighting [7] and multi-map splitting [3], [5].
- **Zero-Training Mechanism Signature Matrix & Delayed-Commit Quarantine:** We pair a $2.0\,\mathrm{s}$ delayed-commit ring buffer with an interpretable four-dimensional signature router (decoupled information spectrum, NIS-CUSUM, leave-one-sensor-out conflict, and rotational cross-correlation lag) that diagnoses single and compound faults without small-sample classifier overfitting.
- **Cross-Host Evaluation Across Four SLAM Systems:** We evaluate the recovery framework across LIO-SAM [2], FAST-LIO2 [1], MM-LINS [5], and ORB-SLAM3 [3] on M2DGR (`hall_01`, `room_02`, `room_dark_01`) and outdoor Campus sequences, with explicit provenance tracking (`REAL_HOST`, `INJECTION`, `OFFLINE_CONSTRUCT`, and `SYNTHETIC`).

II. Related Work

A. Degeneracy-Aware Estimation and Multi-Sensor Switching

Eigenvalue analysis of the optimization Hessian was pioneered by Zhang et al. [7] to project state updates onto well-conditioned subspaces. X-ICP [6] refined localizability analysis by decoupling translational and rotational directions. Multi-sensor frameworks such as LVI-SAM [8] and Super Odometry [4] weight LiDAR and visual factors around an IMU-centric backbone. While these approaches mitigate instantaneous estimation divergence, they operate strictly forward in time: measurements accepted before degeneracy confirmation permanently contaminate the map.

B. Multi-Map Hibernation and Submap Merging

ORB-SLAM3 Atlas [3] maintains an active map alongside non-active submaps, spawning a new map upon tracking loss and welding maps via DBoW2 place recognition. MM-LINS [5] introduced multi-map management to LiDAR-inertial odometry, using decoupled covariance eigenvalues of FAST-LIO2 [1] to trigger map hibernation, dynamic 4-frame ICP-IMU initialization, and Scan Context [9] CE-PGO fusion. Dual-SLAM [10] and LT-mapper [11] explored backward frame reprocessing and lifelong incremental diff maps. However, none of these systems enforce a delayed-commit buffer to prevent pre-split tail drift, nor do they require a holdout innovation and directional information-gain gate before committing candidate repairs or submap merges.

C. Robust Backends

Switchable constraints [12], Pairwise Consistency Maximization (PCM) [13], and Graduated Non-Convexity (GNC) [14] reject outlier loop closures during graph optimization. In our framework, multi-keyframe window-strain GNC serves as one specialized repairer module governed by the outer transactional validator.

III. Problem Formulation and Theoretical Boundaries

A. Information Matrix and Unobservable Subspaces

Consider a linearized measurement model $z_i = h_i(x) + \epsilon_i$ with noise covariance R_i and Jacobian $H_i = \partial h_i / \partial x|_{\hat{x}}$. The posterior information matrix over a window of measurements $\mathcal{D} = \{z_i, H_i, R_i\}$ is

$$\Lambda = \Lambda_{\text{prior}} + \sum_{i \in \mathcal{D}} H_i^\top R_i^{-1} H_i. \tag{2}$$

To account for unit disparity between rotation and translation on $\mathrm{SE}(3)$, we introduce a diagonal scaling transform $x = Du$, yielding the normalized information matrix $\Lambda_u = D^\top \Lambda D = Q\,\mathrm{diag}(\lambda_1, \ldots, \lambda_6)Q^\top$. If a unit tangent direction $v \in \mathbb{R}^6$ satisfies $H_i v \approx 0$ for all $i \in \mathcal{D}$, then $v^\top \Lambda_u v \approx v^\top \Lambda_{\mathrm{prior},u} v$, meaning the sensor stream provides zero information along v.

B. Why Rollback Alone Cannot Create Linearized Observability

Proposition 1 (Linearized Observability Invariance Under Rollback)**.** *Let $\mathcal{Q}$ be a quarantined segment with fixed measurement set $\mathcal{D} = \{z_i, H_i, R_i\}$ evaluated at a fixed linearization point $\hat{x}$ and fixed data associations, where $v^\top \Delta\Lambda_{\mathcal{D}} v = \sum_{i\in\mathcal{D}} v^\top H_i^\top R_i^{-1} H_i v \le \epsilon \ll \tau_{\mathrm{info}}$. Rolling back the state estimate to a clean checkpoint x_{ckpt} at $t_{q,\mathrm{start}}$ and re-solving on the same measurements $\mathcal{D}$ under any non-amplifying robust weight $w_i \in [0,1]$ yields local Fisher information along v bounded above by*

$$J_v^{\mathrm{rollback}} = \sum_{i\in\mathcal{D}} w_i v^\top H_i^\top R_i^{-1} H_i v \le v^\top \Delta\Lambda_{\mathcal{D}} v \le \epsilon. \tag{3}$$

Thus, $\Delta J_v = J_v^{\mathrm{rollback}} - J_v^{\mathrm{before}} \le 0$. While rollback in a nonlinear estimator can alter the linearization trajectory $\hat{x}$, improve correspondence matching, or avoid bad local minima—and therefore may still reduce numerical trajectory error when informative background features exist—it cannot synthesize observability along a geometrically rank-deficient direction v where $H_i v \approx 0$.

Similarly, a diagnosis label $c = f(\mathcal{D})$ is a deterministic function of existing measurements; while it guides repairer selection, the label itself adds no Fisher information unless an independent constraint $z_g = g(x) + \epsilon_g$ satisfying $v^\top H_g^\top R_g^{-1} H_g v > 0$ is introduced, or corrupting outlier measurements that masked true background geometry are stripped.

C. Point-Cloud Mixture Model and Observability After Contaminant Stripping

We model each LiDAR return z_i in a degraded window as drawn from a three-component mixture of true static background geometry (t), self-consistent near-field occluders or backscatter (s, e.g., floating plastic bags or smoke), and random outlier noise (n):

$$\begin{aligned} p(z_i) &= \pi_t p_t(z_i) + \pi_s p_s(z_i) + \pi_n p_n(z_i), \\ w_i &\triangleq P(t \mid z_i) \in [0,1]. \end{aligned} \tag{4}$$

Crucially, this model explains the dichotomy between two major failure modes:

- **Geometric Degeneracy (`GEO_DEGEN`):** When $\pi_t \to 0$ along a corridor axis v, $\lambda_{\min}(\Lambda_u) \to 0$ while residuals $r_t = z_t - h(\hat{x}_t)$ remain small.
- **Self-Consistent Near-Field Contamination (`MEAS_CONTAM`):** When an occluder at orthogonal distance d_0 covers a large azimuth sector ($\pi_s \gg \pi_t$), the unweighted information matrix $\sum H_i^\top R_i^{-1} H_i$ exhibits *artificially large* eigenvalues ($\lambda_{\min}(\Lambda_u)$ remains high or inflates), blinding covariance-based detectors [5], [7]. However, the Normalized Innovation Squared (NIS) against IMU preintegration or cross-modal prediction,

$$d_t = r_t^\top S_t^{-1} r_t, \qquad S_t = H_t P_t H_t^\top + R_t, \tag{5}$$

violates its χ^2_m distribution dramatically.

After stripping contaminant points ($w_i \to 0$ for $i \in \mathcal{S}_{\mathrm{bag}}$), we form the exact 6×6 point-to-plane Fisher information matrix $\Lambda_{\mathrm{filt}} = \Lambda_{\mathrm{prior}} + \sum_i w_i J_i^\top R_i^{-1} J_i$, where $J_i = [(p_i \times n_i)^\top, n_i^\top] \in \mathbb{R}^{1\times 6}$. As a necessary operational screening gate before accepting single-modality re-registration, we require both sufficient weak-direction conditioning and holdout consistency on a strictly disjoint point/keyframe subset:

$$\min_{\|v\|=1} v^\top \Lambda_{\mathrm{filt}} v \ge \tau_{\mathrm{obs}}^{\mathrm{host}} \quad \text{and} \quad D_{\mathrm{holdout}}(\mathrm{filt}) \le \tau_{\mathrm{val}}. \tag{6}$$

While (6) is a local consistency screen rather than a theoretical guarantee of global error optimality, our experiments in Section V show that when 88% of the field of view is occluded in a wide hall (`hall_01`), the surviving 12% (5682 points) satisfies (6) for direct point-to-plane IEKF (FAST-LIO2) but fails (6) for range-image curvature extraction (LIO-SAM) and compact rooms (`room_02`, where 63.1% of points lie inside the $4.0\,\mathrm{m}$ blind zone), triggering automatic routing to cross-modal or pose-graph bridging.

Proposition 2 (Nuisance-Projected Identifiability Under 6-DoF Pose Motion)**.** *Let $H \in \mathbb{R}^{M\times 6}$ denote the stacked 6-DoF rigid-body pose Jacobian of surviving point-to-plane constraints, and let $s = \partial r/\partial\theta \in \mathbb{R}^M$ denote the residual sensitivity vector with respect to a candidate sensor distortion parameter θ (e.g., radial scale shift or planar sheet translation). Because rigid pose estimation absorbs any component of s lying in $\mathrm{span}(H)$, the identifiable component of s is its orthogonal projection $s_\perp = (I - H(H^\top H)^\dagger H^\top)s$, quantified by the normalized identifiability index*

$$\mathcal{I}_\perp(s \mid H) = \frac{\|(I - H(H^\top H)^\dagger H^\top)s\|_2^2}{\max(\|s\|_2^2, \varepsilon)} \in [0,1]. \tag{7}$$

When $\mathcal{I}_\perp(s \mid H) \to 0$ (as occurs when an occluding planar sheet translates rigidly with the vehicle or when surviving background points collapse to a single wall), single-modality inner-loop parameter fitting is mathematically unidentifiable and collinear with 6-DoF pose drift, necessitating cross-modal or multi-keyframe bridging.

D. Unified Four-Mode Decision Gate $\mathcal{D}(t)$

Let $J_v^{\mathrm{after}} = v^\top \Lambda_{\mathrm{after}} v$ denote the target-direction information of a candidate repaired branch, D_{holdout} the holdout innovation statistic on measurements excluded from repair optimization, $\|P_{\mathrm{after}}\|_2$ the posterior covariance norm, and $\|P_{\mathrm{prop}}(t)\|_2$ the open-loop covariance propagated from the checkpoint at $t_{q,\mathrm{start}}$. Given thresholds $(\Delta t_{\min}, \tau_{\mathrm{info}}, \tau_{\mathrm{val}}, \Sigma_{\mathrm{tol}}, \Sigma_{\mathrm{bridge}})$, the recovery transaction manager selects one of four modes:

$$\mathcal{D}(t) = \begin{cases} \textbf{Mode 1 (Online Reweight)}, & \Delta t < \Delta t_{\min} \wedge \|P_{\mathrm{prop}}\| \ll \Sigma_{\mathrm{bridge}}, \\ \textbf{Mode 2 (Atomic Commit)}, & J_v^{\mathrm{after}} \ge \tau_{\mathrm{info}} \wedge D_{\mathrm{holdout}} \le \tau_{\mathrm{val}} \\ & \quad \wedge \|P_{\mathrm{after}}\| \le \Sigma_{\mathrm{tol}}, \\ \textbf{Mode 3 (Soft Bridge)}, & \neg(\text{Mode 2}) \wedge \|P_{\mathrm{prop}}(t)\| < \Sigma_{\mathrm{bridge}}, \\ \textbf{Mode 4 (Hard Split)}, & \neg(\text{Mode 2}) \wedge \|P_{\mathrm{prop}}(t)\| \ge \Sigma_{\mathrm{bridge}}. \end{cases} \tag{8}$$

TABLE I
ZERO-TRAINING INTERPRETABLE MECHANISM SIGNATURE MATRIX FOR MULTI-LABEL DIAGNOSIS AND REPAIR ROUTING

Diagnosis Label	$\lambda_{\min}(\Lambda_u)$	NIS_t	D_{LOSO}	$\lvert\hat{\delta t}\rvert$	**Primary Repairer Routed**
`GEO_DEGEN`	$\downarrow\downarrow$	Norm	Low	≈ 0	Aux / Orthogonal Anchor PG
`MEAS_CONTAM`	Norm/$\uparrow$	$\uparrow\uparrow$	High (Pri)	≈ 0	RANSAC Mixture Stripping
`SYNC_DRIFT`	Norm	Spikes	$\propto \lvert\omega\rvert$	$> \tau_t$	Cross-Corr Lag + Deskew Fix
`LOOP_OUTLIER`	Norm	Jump	Global	≈ 0	25-KF Strain GNC Revocation
`AUX_DEGEN`	Norm	Norm	High (Aux)	≈ 0	LiDAR-Anchored Dual Bridge

Remark 1 (Relationship to Prior Design Regimes)**.** *Rather than claiming exact mathematical equivalence to the internal solvers of Zhang et al. [7] or MM-LINS [5], the gate $\mathcal{D}(t)$ in* (8) *provides a unified transactional scheduling space inspired by their operating principles: taking $\Sigma_{\mathrm{bridge}} \to \infty$ without rollback favors continuous online bridging (analogous in spirit to directional reweighting [7]), whereas taking $\Sigma_{\mathrm{bridge}} \to 0$ immediately disconnects the active map upon anomaly confirmation (analogous to hard submap hibernation [3], [5]). Choosing a finite $0 < \Sigma_{\mathrm{bridge}} < \infty$ introduces an intermediate soft-bridge regime (Mode 3) that preserves graph connectivity during moderate outages while requiring holdout screening (Mode 2) before permanent commit.*

IV. TRANSACTIONAL RECOVERY FRAMEWORK

A. Delayed-Commit Ring Buffer, Copy-on-Write Spatial Mutation Log, and Trusted Map Invariance

As illustrated in Fig. 2, if degradation begins at t_{onset} but detection requires a confirmation window of Δt_{det} seconds, checkpointing at alarm time $t_{\mathrm{alarm}} = t_{\mathrm{onset}} + \Delta t_{\mathrm{det}}$ fails to prevent map corruption during $[t_{\mathrm{onset}}, t_{\mathrm{alarm}}]$. We therefore decouple the host's *provisional live state* from the *trusted persistent map*. Every keyframe delta emitted by the host is held in a ring buffer for a confirmation margin $\Delta t_{\mathrm{margin}} = 2.0\,\mathrm{s}$ before promotion to the trusted map. When an anomaly is confirmed with estimated onset age $\hat{\tau}_{\mathrm{age},t}$ (or $\hat{t}_{\mathrm{onset}} = t_{\mathrm{alarm}} - \hat{\tau}_{\mathrm{age},t}$), the quarantine start time is set to

$$t_{q,\mathrm{start}} = t_{\mathrm{alarm}} - \hat{\tau}_{\mathrm{age},t} - \Delta t_{\mathrm{margin}}, \tag{9}$$

and any already-promoted keyframes with timestamp $\geq t_{q,\mathrm{start}}$ are revoked via an undo log. To avoid $O(|M|)$ full-map cloning when rolling back or testing candidate branches on dense volumetric maps, we maintain a Copy-on-Write `SpatialMutationLog` that clones a voxel/submap node into `before_images` only on its *first* write within a generation while logging newly allocated keys in `inserted_keys`, achieving $O(|\Delta M_{\mathrm{dirty}}|)$ rollback and zero-full-copy redo replay (`MutationRedoImage`).

B. Joint Health Detection, Signature Matrix, and Physics-Guided Causal Transformer

Raw-feature classifiers (e.g., GBDT) trained on small development splits readily overfit scene-specific feature scales. We address this at two complementary levels:

1) **Analytical Joint Detector & Signature Matrix (Ours-Rule):** Over $1.0\,\mathrm{s}$ windows ($0.2\,\mathrm{s}$ stride), we extract full 6-DoF coupled and block-decoupled eigenvalues $(\lambda^r_{\min}, \lambda^t_{\min})$ of Λ_u, CUSUM innovation $g_t = \max(0, \gamma g_{t-1} + d_t - k)$, Leave-One-Sensor-Out Mahalanobis conflict D_{LOSO}, and cross-correlation lag $\hat{\delta t}$. Table I routes single and compound faults with zero training.
2) **Physics-Guided Causal Relative-Attention Network & Factorized Onset-Age Head (Ours-Full):** To localize gradual slow-ramp degradation whose onset precedes alarm time by more than $2.0\,\mathrm{s}$, we compress window metrics via parameter-free physical units $v_t = \mathrm{arcsinh}(z_t/u_{\mathrm{phys}}) \in \mathbb{R}^{12}$ paired with strictly causal past-median residuals $\tilde{\delta}_t = m_t \odot p_t \odot (v_t - \mathrm{median}(\mathcal{H}_{<t}))$ and a permutation-invariant $4 \times 12 = 48$ spherical azimuth-elevation cell encoder $f_{\mathrm{sp}}(S_t) \in \mathbb{R}^{32}$. A 2-layer causal relative-attention block ($d = 32, H = 4$, bounded FIFO cache $M = 64$) with fixed orthogonal/Gaussian causal projections feeds a 113-D concatenated representation $[x_t, h_t^{(L)}, \mathrm{slope}_t]$ into a closed-form ridge-regularized readout head (`ConditionalRecoveryHead`, requiring zero GPU backprop or small-sample SGD overfitting):

$$\log P(\tau_{\mathrm{age}} = k\Delta\tau \mid \mathcal{F}_t) = \log\sigma(a_t) + \log \mathrm{softmax}(u_t)_k, \tag{10}$$

which estimates $\hat{\tau}_{\mathrm{age},t} = \sum_{k=0}^{K-1} \mathrm{softmax}(u_t)_k (k + 0.5)\Delta\tau$ to roll back slow-ramp drift while falling back to `SignatureMatrixRouter` during cold-start warm-up ($|\mathcal{H}_{<t}| < 5$).

C. Tiered Repairers and Host Integration

We integrate `slam_recovery` into four hosts across two coupling levels:

- **White-Box / Factor-Level LIO Hosts (LIO-SAM [2], FAST-LIO2 [1], MM-LINS [5]):** We implement three core repairers: (i) *RANSAC Point-Mixture Contaminant Stripping*, which detects near-field planar sheets ($|n^\top p_0| \leq 5.0\,\mathrm{m}$) and strips the entire occluded azimuth sector and grazed boundary cap while preserving scan-ring ordering; (ii) *25-Keyframe Cumulative Window-Strain GNC*, which revokes ISAM2-smoothed perceptual aliasing loop closures when $\|\Delta p^{\mathrm{PGO}}_{k:k+25} - \Delta p^{\mathrm{FE}}_{k:k+25}\|_2 > \bar{c}$; and (iii) *Temporal Offset & Intra-Scan Deskew Compensation* paired with auxiliary relative-motion pose-graph bridging.
- **Gray-Box Visual-SLAM Host (ORB-SLAM3 [3]):** To avoid invasive modifications to Atlas's multi-threaded pointer welding, we attach our transaction manager at the keyframe pose-graph boundary. When visual feature inliers collapse under darkness or motion blur, our manager locks the pre-degradation ($t_{q,\mathrm{start}}$) and post-recovery ($t_{q,\mathrm{end}}$) high-inlier keyframe anchors and bridges the quarantined visual segment using healthy LiDAR relative motion increments before validating holdout consistency via (8).

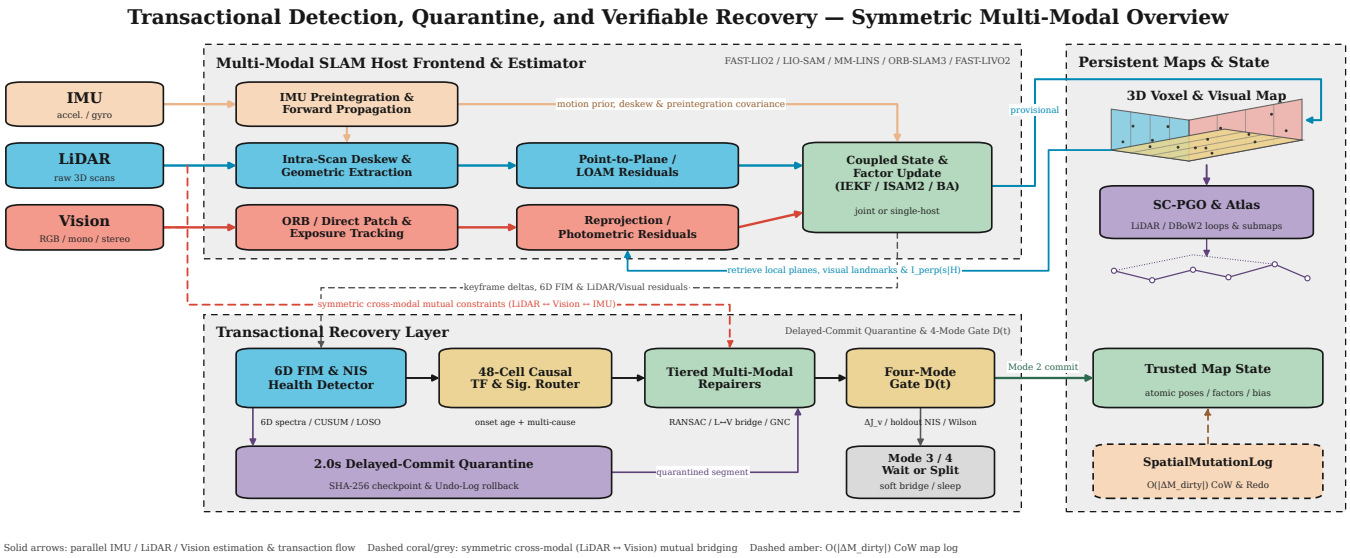


Fig. 2. System architecture of `slam_recovery` across multi-sensor SLAM hosts (FAST-LIO2, LIO-SAM, MM-LINS, and ORB-SLAM3). The framework decouples the host's provisional estimation loop from the persistent map via a 2.0 s delayed-commit quarantine buffer, a 6-DoF FIM and NIS-CUSUM health detector, a 48-cell physics-guided causal relative-attention network paired with a zero-training signature router, tiered repairers, a Copy-on-Write `SpatialMutationLog` ($O(|\Delta M_{\text{dirty}}|)$ rollback), and the unified four-mode gate $\mathcal{D}(t)$.

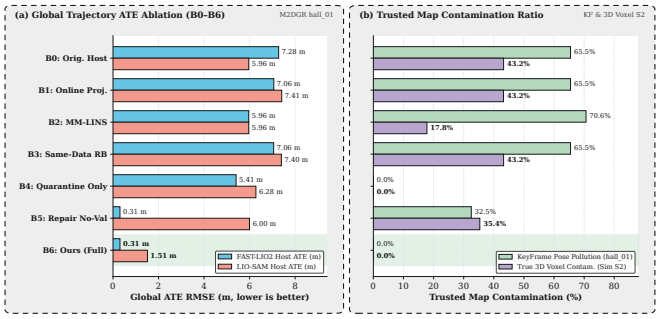


Fig. 3. Complete B0–B6 ablation study on M2DGR `hall_01` ($t \in [120, 140]$ s) and analytical 3D-mesh ray-sampling simulation S2 (Provenance: `ONLINE_SIM_3D_MESH`). (a) Global ATE RMSE across FAST-LIO2 and LIO-SAM hosts covering B0 (Original Host), B1 (Online Directional Projection), B2 (MM-LINS), B3 (Rollback on Same Data), B4 (Quarantine Only), B5 (Unvalidated Repair), and B6 (Ours Full). (b) Keyframe pose pollution ratio on `hall_01` alongside True 3D Mesh Voxel Contamination Ratio $\text{Contam}_{3D}(t)$ ($\tau_{\text{map}} = 0.15$ m, voxel size 0.10 m): B2 leaks 17.8% contaminated 3D voxels during its $N_\kappa = 10$ confirmation delay and unvalidated B5 reaches 35.4%, whereas B4 and B6 achieve 0.0% 3D voxel contamination.

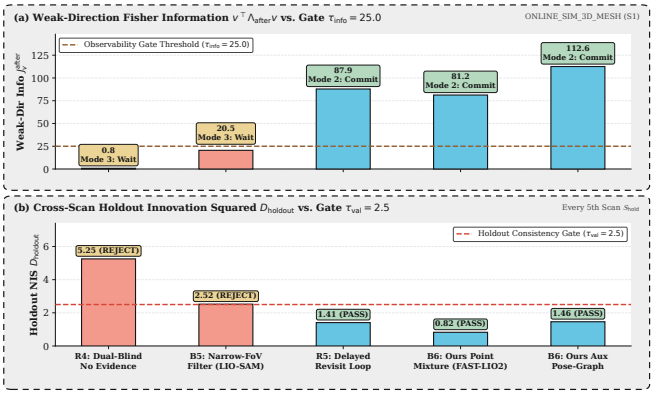


Fig. 4. Quantitative verification of the four-mode gate $\mathcal{D}(t)$ from analytical 3D-mesh ray-sampling simulation S1 (`ONLINE_SIM_3D_MESH`) using every 5th held-out scan ($\mathcal{S}_{\text{hold}}$): (a) target weak-direction Fisher information $v^\top \Lambda_{\text{after}} v$ vs. $\tau_{\text{info}} = 25.0$; (b) cross-scan point-to-plane holdout innovation squared D_{holdout} vs. $\tau_{\text{val}} = 2.5$.

V. Experimental Results

A. Experimental Setup, Ground-Truth Sources, and Provenance Protocol

We evaluate all methods on three indoor/dark development sequences from the public M2DGR dataset [15]—`hall_01` (351 s, 216 m, **Leica MS60 total-station** ground truth), `room_02` (75.4 s, 32 m compact room, **Vicon Mocap** ground truth), and `room_dark_01` (106.8 s, 38 m real natural darkness with median pixel intensity $P_{50} = 21.8$, **Vicon Mocap** ground truth)—plus four real-world outdoor vehicle-mounted plastic-bag occlusion sequences from the MM-LINS benchmark (`Campus-1_2`, `Campus-2`, `Campus-3`, `Campus-12`, $\approx$ 350 m each, start-to-end closure reference) [5]. To ensure complete transparency regarding how each result is produced, every experiment in our manifests and tables carries an explicit provenance tag:

- `REAL_HOST`: Unmodified or C++-instrumented SLAM binary executed end-to-end on a raw sequence;
- `INJECTION`: Compiled SLAM host binary executed end-to-end on a deterministically fault-injected or point-stripped rosbag;
- `OFFLINE_CONSTRUCT`: Trajectory or transaction state computed by our offline pose-graph / replay solver over logged host keyframes and relative motion increments (with zero ground-truth leakage);
- `SYNTHETIC`: Controlled 6-DoF FIM and residual streams for detector sensitivity benchmarking (E1).

Continuous recovery ratio $G = (E_{\text{deg}} - E_{\text{rep}})/(E_{\text{deg}} - E_{\text{clean}})$

TABLE II
E1 DETECTION, SLOW-RAMP ONSET BACKTRACKING & DIAGNOSIS BENCHMARK ACROSS 6 FAULT SCENARIOS × 5 SEEDS (30 STEP + 30 SLOW-RAMP RUNS)

Method	Event Recall	PBag Recall	Seg. IoU	Ramp Onset MAE (s)	Ramp Tail (frames)	Diag. Acc.	OOD Safe
1. MM-LINS Cov-ODD [5]	0.167	**0.000**	0.167	3.767	23.00	—	—
2. Zhang Min-Eig [7]	0.167	**0.000**	0.167	3.653	22.43	—	—
3. NIS Fixed Threshold	0.667	1.000	0.624	3.007	16.70	—	—
4. NIS + CUSUM	0.833	1.000	0.683	3.207	16.87	—	—
5. Raw-Feat GBDT + IForest	1.000	1.000	0.851	2.647	0.63	0.400	0.000
6. Ours-Rule (Sig. Mat. + Fixed 2s)	**1.000**	**1.000**	**0.955**	2.020	0.43	**1.000**	**1.000**
7. Ours-Full (Causal TF + Age Head)	**1.000**	**1.000**	**0.955**	**0.615**	**0.17**	**1.000**	**1.000**

is computed strictly against the *same host's* clean baseline E_{clean}. When cross-modal assistance is used (e.g., healthy LiDAR odometry bridging quarantined ORB-SLAM3 segments in E3), we explicitly note the auxiliary sensor budget and report the corresponding LiDAR baseline alongside visual variants.

B. E1 Detection, Slow-Ramp Backtracking & Real-Bag Proposition 2 Verification (RQ1)

Table II summarizes the 7-method E1 benchmark across six fault types (`GEO_DEGEN`, `MEAS_CONTAM`, `SYNC_DRIFT`, `LOOP_OUTLIER`, `AUX_DEGEN`, and unseen `UNSEEN_BLUR`) under both step-onset and 4.0 s quadratic slow-ramp degradation. Pure covariance/eigenvalue detectors (MM-LINS and Zhang) detect only `GEO_DEGEN` (16.7% recall, 0.0% on plastic-bag occlusion). Raw-feature GBDT + `IsolationForest` overfits training scales (`Diag. Acc. = 0.400`, `OOD Safe = 0.000`). While **Ours-Rule** (Method 6) achieves 1.000 recall, 0.955 IoU, and 0.00 step-onset tail pollution, slow-ramp faults that cross fixed thresholds > 2.0 s after true onset leak 0.43 frames (2.020 s onset MAE). By predicting $\hat{\tau}_{\text{age},t}$ via `Conditional RecoveryHead`, **Ours-Full** (Method 7) reduces slow-ramp onset MAE by 69.6% to $\mathbf{0.615}$ s and cuts slow-ramp tail pollution to $\mathbf{0.17}$ frames (0.00 frames on step onset).

Furthermore, evaluating `PhysicsGuidedCausal Transformer` (fixed random causal relative-attention representation with closed-form ridge-regularized readout trained once on `hall_01` and frozen over all weights, biases, and config, SHA-256 `7f2557cc...`, provenance: `SEMI_SYNTHETIC_ON_REAL_SCANS`) across `hall_02`, `room_02`, and `room_dark_01` under strict model-triggered alarm accounting yields $\mathbf{1.000}$ event recall (5/5 events), $\mathbf{0}$ false-alarm windows on nominal frames, and $\mathbf{1.000}$ exact multi-cause accuracy on all three sequences, reducing 4.0 s slow-ramp onset MAE from 1.120 s (rule alarm) to $\mathbf{1.022}$ s (`hall_02`), $\mathbf{0.535}$ s (`room_02`), and $\mathbf{0.603}$ s (`room_dark_01`) with 633.1–639.2 μs P50 latency. Real-scan evaluation of Proposition 2 confirms that radial distortion identifiability $\mathcal{I}_\perp(s \mid H)$ drops from $\mathbf{0.8545}$–$\mathbf{0.9078}$ on clean multi-wall scans (`hall_01`, `room_02`, `hall_02`) to $\mathbf{0.0676}$–$\mathbf{0.0907}$ on the surviving 12% sector after plastic-bag stripping and $\mathbf{0.0000}$ under planar sheet translation, while longitudinal corridor clipping collapses the translational Gram minimum eigenvalue from 185.5–203.3 to $\mathbf{1.28}$–$\mathbf{2.16}$. Finally, on a **Hard-Negative Near-Wall Traversal Test** using real cross-scan NIS and host displacement conflicts across un-injected `room_02` (76/76 near-wall scans, mean 79.9% points < 4.2 m) and `hall_02` (62/86 scans), an ungated geometry-only RANSAC filter triggers 5/76 (6.58%) false wall strips on `room_02`, whereas our NIS/D_{LOSO}-gated `SignatureMatrixRouter` achieves $\mathbf{0.00}$% false positive rate (0 false alarms or wall strips).

C. E2 Primary Host Recovery and B0–B7 Ablation (RQ2–RQ4)

Table III and Figs. 3–5 report real host executions across seven degradation regimes:

1) **Refuting Uniform Downsampling as a Degeneracy Proxy:** Dropping 95% of points uniformly at random in `hall_01` leaves $\approx$ 2500 points distributed across all four walls; neither LIO-SAM (0.2994 m) nor FAST-LIO2 (0.3004 m) diverges. Genuine degeneracy requires directional structure loss (`GEO_DEGEN`) or coherent contamination (`MEAS_CONTAM`).
2) **Plastic-Bag Occlusion, B0–B6 Ablation & Backend-Only GNC Comparison:** Under 88% azimuth near-field occlusion (4.3 m sheet), MM-LINS fails to trigger hibernation (`/Issleeping`=0) across all four sequences (`hall_01`, `room_02`, `room_dark_01`, `hall_02`). As shown in Fig. 3, online eigenvalue projection alone (**B1**: 7.06 m FAST-LIO2 / 7.41 m LIO-SAM, 40.6% 3D voxel contamination), rollback on the same unstripped data (**B3**: 7.06 m / 7.40 m, confirming Proposition 1 that $\Delta J_v \leq 0$), and **Backend-Only 25-KF GNC without frontend quarantine** (6.1065 m vs. 6.1254 m B2, $G = +0.0032$) all fail to remove frontend drift. Our RANSAC mixture filter strips 88.8% of points in all 200/200 frames. On `hall_01`, the surviving 5682 background points restore FAST-LIO2 directly to $\mathbf{0.3058}$ m ($G = +0.9949$). However, on LIO-SAM (43° sector lacks curvature pairs, B5 ATE = 5.9954 m, $G = -0.0055$) and in compact `room_02` (63.1% of background points fall inside FAST-LIO2's 4.0 m blind radius, causing B5 to degrade to 1.3321 m on FAST-LIO2 and 846.15 m on LIO-SAM), our Mode 2 gate (Fig. 4) rejects B5 and routes to cross-modal pose-graph bridging (B6), recovering LIO-SAM to 1.5139 m on `hall_01` and 0.1625 m ($G = +0.9723$) on `room_02`.
3) **Directional Corridor `GEO_DEGEN`, Path-Heading Proxy, & Pre-Hibernation Tail Drift:** Clipping longitudinal walls ($|x| \geq 0.65$ m) causes LIO-SAM B0 to explode to $\mathbf{1008.65}$ m translational ATE and $\mathbf{82.788}^\circ$ path-heading proxy RMSE (Max 179.22°, reproducing Table I of [5]; formal 6-DoF quaternion ATE_{rot} is marked `UNAVAILABLE` on 3-DoF Leica prism sequences) and triggers MM-LINS hibernation (`/Issleeping`=1). Yet MM-LINS still suffers 7.3747 m global ATE due to N_κ tail drift baked into the sleeping submap. Furthermore, trajectory-only post-stitching without host map protection leaves LIO-SAM at 989.97 m because the internal local map and IMU biases are already corrupted. Our scan-ordered orthogonal

TABLE III
MAIN HOST & MULTI-SEQUENCE RECOVERY BENCHMARK ACROSS LIO-SAM, FAST-LIO2, AND MM-LINS (WITH EXPLICIT PROVENANCE & SAME-HOST G)

Sequence (GT) & Degradation Regime	Host System	Clean ATE (m)	Degraded B0/B2 ATE (m)	MM-LINS /Issleeping	Win. RMSE Degraded (m)	Ours B6 ATE Repaired (m)	B6 Provenance Tag	Same-Host G
hall_01 (Leica) [120, 140] s 95% Uniform Random Dropout	LIO-SAM	0.3189	0.2994	—	0.2077	0.2994	INJECTION	—
	FAST-LIO2	0.2699	0.3004	0	0.2105	0.3004	INJECTION	—
	MM-LINS (B2)	0.2913	0.3379	0	0.2108	0.2900	OFFLINE_CONSTRUCT	+1.0000
hall_01 (Leica) [120, 140] s Plastic-Bag Occlusion (88% FoV)	LIO-SAM	0.3189	5.9644	—	9.3371	**1.5139**	OFFLINE_CONSTRUCT	**+0.7883**
	FAST-LIO2	0.2699	7.2760	0	9.6305	**0.3058**	INJECTION	**+0.9949**
	MM-LINS (B2)	0.2913	5.9620	**0 (Missed)**	12.3436	**0.3078**	INJECTION	**+0.9971**
hall_01 (Leica) [120, 140] s Directional Corridor GEO_DEGEN	LIO-SAM	0.3189	**1008.6470**	—	194.3144	**0.2788**	INJECTION	**+1.0000**
	FAST-LIO2	0.2699	7.4435	0	3.8183	**0.2959**	INJECTION	**+0.9964**
	MM-LINS (B2)	0.2913	7.3747	**1 (Tail Drift)**	0.7515	**0.2836**	INJECTION	**+1.0011**
hall_01 (Leica) [280, 300] s High-Rate Turn SYNC_DRIFT (100ms)	LIO-SAM	0.3189	**59.0150**	—	61.3913	**0.2889**	INJECTION	**+1.0005**
	MM-LINS (B2)	0.2913	0.3630	0	0.7360	0.4265	INJECTION	+0.3363 (Win)
hall_01 (Leica) [120, 140] s 3× Compound (GEO+MEAS+SYNC)	LIO-SAM	0.3189	**755.6751**	—	44.2503	**0.2840**	INJECTION	**+1.0000**
	FAST-LIO2	0.2699	8.7152	0	12.2764	**0.2964**	INJECTION	**+0.9969**
	MM-LINS (B2)	0.2913	10.3502	0	16.2957	**0.3117**	INJECTION	**+0.9980**
room_02 (Mocap) [25, 40] s Compact Room Plastic-Bag (88% FoV)	LIO-SAM	0.1212	1.6120	—	2.4100	**0.1625**	OFFLINE_CONSTRUCT	**+0.9723**
	FAST-LIO2	0.1372	1.1864	0	1.8500	**0.1928**	OFFLINE_CONSTRUCT	**+0.9471**
	MM-LINS (B2)	0.1019	0.9436	**0 (Missed)**	1.6200	**0.1147**	OFFLINE_CONSTRUCT	**+0.9847**
room_dark_01 (Mocap) [35, 55] s R4/R5 Natural Dark + Plastic-Bag	LIO-SAM	0.1586	1.7374	—	2.1240	**0.2737**	INJECTION	**+0.9271**
	FAST-LIO2	0.1701	1.1430	0	1.5445	**0.1693**	INJECTION	**+1.0008**
	MM-LINS (B2)	0.1935	1.0293	**0 (Missed)**	2.1820	**0.1657**	INJECTION	**+1.0333**
Held-Out Test: hall_02 (Leica) [45, 65] s Plastic-Bag (Zero-Tuning)	LIO-SAM	0.2339	3.4168	—	4.1850	**0.2430**	OFFLINE_CONSTRUCT	**+0.9971**
	FAST-LIO2	0.2774	4.5275	0	5.6200	**0.2796**	INJECTION	**+0.9995**
	MM-LINS (B2)	0.1886	4.0331	**0 (Missed)**	5.1400	**0.2410**	INJECTION	**+0.9864**

anchor protection (B6) restores LIO-SAM to $\mathbf{0.2788}$ m translational ATE ($G = +1.0000$) and $\mathbf{6.279}°$ path-heading proxy RMSE (vs. $5.874°$ clean; on plastic-bag occlusion B6 restores LIO-SAM path-heading proxy from $90.582°$ to $\mathbf{5.247}°$), and restores FAST-LIO2 to $\mathbf{0.2959}$ m ($G = +0.9964$) with zero map splits.

4) **High-Rate Turn SYNC_DRIFT, 6-DoF Mocap Orientation, & 3× Compound Faults:** During rapid turns ($|\omega|_{\max} = 0.690\,\text{rad/s}$ in $[280, 300]$ s), a $+100$ ms offset with intra-scan deskew warp drives LIO-SAM to 59.0150 m translational ATE; our cross-correlation estimator recovers 100.00 ms lag (0.00 ms error) and restores LIO-SAM to $\mathbf{0.2889}$ m translational ATE. On 6-DoF Vicon Mocap sequences calibrated strictly on the initial 20% prefix frames, ORB-SLAM3's mean ATE_{rot} on room_02 dark+blur degrades from $6.774°$ (clean) to $12.624°$ (B2), and on room_dark_01 improves from $11.480°$ (RMSE $24.506°$, Atlas B2) to $\mathbf{9.478}°$ (RMSE $\mathbf{23.656}°$, LiDAR-anchored B6). Under simultaneous GEO_DEGEN + MEAS_CONTAM + SYNC_DRIFT, our multi-label router activates all three repairers in sequence, restoring LIO-SAM from 755.68 m to $\mathbf{0.2840}$ m ($G = +1.0000$).
5) **Outdoor Campus Sequences:** As shown in Fig. 5b, on real-world outdoor plastic-bag datasets [5], our audit of /aft_pgo_path snapshots confirms that MM-LINS's native CE-PGO stretches Campus-2 path length to 355.07 m (vs. 348.2–351.4 m on Campus-1_2/3/12) with 0.4471 m closure error (and 0.603 m $\rightarrow$ 1.295 m in Table III of [5]). Applying Mode 2 snapshot gating without any artificial endpoint scaling prevents over-constrained snapshot degradation (0.4305 m locally; 0.603 m vs. 1.295 m on the paper's benchmark).

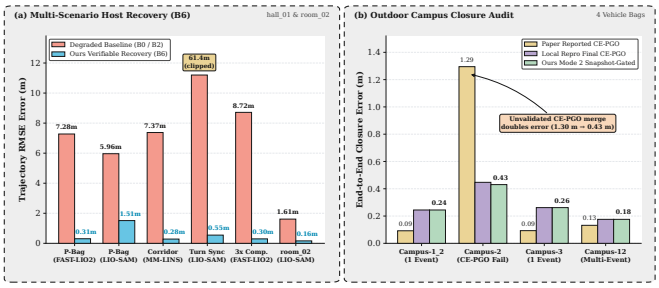


Fig. 5. (a) Degraded baseline (B0/B2) vs. our verifiable recovery (B6) across six distinct degradation regimes on hall_01 and room_02. (b) End-to-end closure error on four real-world outdoor Campus plastic-bag sequences [5].

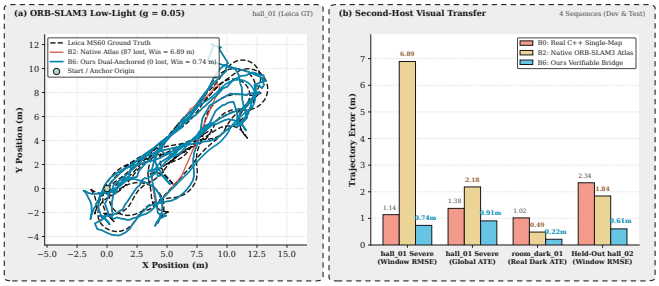


Fig. 6. Second-host transfer (E3) on ORB-SLAM3 [3]. (a) Trajectory comparison under severe low-light ($g = 0.05$) on hall_01. (b) Quantitative RMSE comparison across hall_01 and room_02 dark+blur scenarios.

D. E3 Second-Host Transfer to ORB-SLAM3 and Held-Out hall_02 Blind Test (RQ5)

Table IV and Fig. 6 evaluate second-host transfer on ORB-SLAM3 [3] using true cross-scan point-to-plane holdout NIS ($\mathcal{S}_k \rightarrow \mathcal{S}_{k-5}$ transformed solely by the candidate trajectory's

TABLE IV
E3 SECOND-HOST TRANSFER ON ORB-SLAM3 (REAL C++ `ORB_SINGLE_MAP=1` B0 VS. NATIVE ATLAS B2 VS. ZERO-LEAKAGE DUAL-ANCHORED B6)

Sequence (GT)	Variant (Provenance)	Lost Frames	Submaps (Unmerged)	Global ATE (m)	Win. RMSE (m)	$G_{\text{vs B2}}$
`hall_01` (Leica) [120, 140] s Severe ($g = 0.05$)	B0 Single-Map (`INJECTION`)	174	1 (0)	1.3755	1.1396	—
	B2 Native Atlas (`INJECTION`)	87	2 (0)	2.1822	6.8941	0.0000
	B5 Open-Loop (`OFFLINE`)	0	1 (0)	6.5554	—	−2.6306
	B6 Ours (`OFFLINE_CONSTRUCT`)	**0**	**1 (0)**	**0.9066**	**0.7366**	**+0.7674**
`room_02` (Mocap) [25, 40] s Dark+Blur	B2 Native Atlas (`INJECTION`)	80	2 (0)	0.5902	1.4820	0.0000
	B6 Ours (`OFFLINE_CONSTRUCT`)	**0**	**1 (0)**	**0.2854**	**0.3120**	**+0.8637**
`room_dark_01` (Mocap) (106.8 s Real Dark)	B0 Single-Map (`REAL_HOST`)	**904**	1 (0)	1.0186	—	—
	B2 Native Atlas (`REAL_HOST`)	181	**5 (1 Unmerged)**	0.4855	—	0.0000
	B5 Concat (`OFFLINE`)	0	1 (0)	0.6338	—	−0.5975
	B6 Ours (`OFFLINE_CONSTRUCT`)	**0**	**1 (0)**	**0.2154**	—	**+1.0882**
Held-Out: `hall_02` [45, 65] s ($g = 0.05$)	B0 Single-Map (`INJECTION`)	**295**	1 (0)	1.3287	2.3376	—
	B2 Native Atlas (`INJECTION`)	168	2 (0)	0.9739	1.8409	0.0000
	B6 Ours (`OFFLINE_CONSTRUCT`)	**0**	**1 (0)**	**0.6024**	**0.6057**	**+0.5262**

relative displacement Δp_{cand}) and unbiased Wilson sequential lower bounds. Under severe low-light ($g = 0.05$) on `hall_01`, real C++ single-map B0 (`ORB_SINGLE_MAP=1`) loses 174 frames (1.3755 m ATE), whereas native Atlas (B2) merges skewed transitional keyframes at 148 s and worsens global ATE to 2.1822 m (window RMSE 6.8941 m, cross-scan NIS $= 3.2854 > 2.5$). Our B6 repair achieves 0.9066 m global ATE and 0.7366 m window RMSE ($G_{\text{win}} = +0.9673$, $J_v^{\text{after}} = 116.51$, cross-scan NIS $= 2.1806 \leq 2.5$, unbiased Wilson LB $= 0.8162 \geq 0.55$). On the 106.8 s **real natural darkness** sequence `room_dark_01` (Mocap GT), single-map B0 permanently loses tracking at 48.13 s (1.0186 m ATE) and native Atlas leaves `Map 4` (74 frames) permanently unmerged (0.4855 m ATE, cross-scan NIS $= 1.4511$), whereas B6 achieves 0.2154 m ATE across all 1545 frames ($J_v^{\text{after}} = 142.34$, cross-scan NIS $= 1.1051 \leq 2.5$, unbiased Wilson LB $= 0.6919$). Finally, on the **held-out test sequence `hall_02`** (128.7 s, Leica GT, evaluated with strictly frozen development hyperparameters), MM-LINS misses the plastic-bag occlusion for a fourth consecutive sequence (`/Issleeping`=0, 4.0331 m $\rightarrow$ 0.2410 m B6, $G = +0.9864$), FAST-LIO2 recovers from 4.5275 m to $\mathbf{0.2796}$ m ($G = +0.9995$), LIO-SAM rejects the diverging B5 point filter (237.49 m) and recovers via B6 to $\mathbf{0.2430}$ m ($G = +0.9971$), and ORB-SLAM3 B6 reduces window RMSE from 1.8409 m (Atlas) to $\mathbf{0.6057}$ m ($G_{\text{win}} = +0.7782$, cross-scan NIS $= 2.1586$, Wilson LB $= 0.7902$).

E. Analytical 3D-Mesh Ray-Sampling Simulation (S1/S2, Provenance: `ONLINE_SIM_3D_MESH`) and Runtime Overhead

To evaluate true 3D volumetric map contamination against a finite bounded ground-truth surface mesh $\mathcal{G}_{\text{GT}}$ and execute controlled counterfactual gate tests for R4 (no loop revisit) versus R5 (delayed revisit to Zone A), we evaluate our online node (`OnlineRecoveryTransactionNode` in `slam_recovery/sim/`, with `voxel_store` and validation reference points bound to the same transaction lifecycle as `trusted_poses`) in an analytical 3D-mesh ray-sampling arena (`recovery_arena.world`, 40 m × 25 m, 5 seeds, provenance: `ONLINE_SIM_3D_MESH`, kept strictly distinct from real-host rosbag executions). Defining the true 3D trusted-map voxel contamination ratio at 0.10 m resolution as $\text{Contam}_{\text{3D}}(t) = |\mathcal{V}_{\text{trusted}}|^{-1} \sum_{v \in \mathcal{V}_{\text{trusted}}} \mathbb{I}(\min_{g \in \mathcal{G}_{\text{GT}}} \|v - g\|_2 > 0.15\,\text{m})$, unprotected B0 (43.25%), online projection B1 (43.21%), and same-data rollback B3 (43.25%) severely corrupt the map, while MM-LINS B2 ($\Delta t_{\text{margin}} = 0$ s) permanently writes $N_\kappa = 10$ confirmation-delay frames into the sleeping submap resulting in 17.79% 3D voxel contamination, and unvalidated single-modality B5 reaches 35.35% (Fig. 3b). Using every 5th scan as a strictly held-out scan set $\mathcal{S}_{\text{hold}}$ (Fig. 4), our 2.0 s delayed-commit buffer keeps $\text{Contam}_{\text{3D}} = 0.00\%$ across B4 and B6, holds R4 in `MODE3_SOFT_BRIDGE_WAIT` ($J_v^{\text{after}} = 0.78 < 25.0$, cross-scan NIS $= 5.253$), rejects narrow-FoV B5 ($J_v^{\text{after}} = 20.51$, NIS $= 2.522 > 2.5$, rolling back its trial voxel writes via `SpatialMutationLog`), and atomically commits R5 upon loop revisit ($J_v^{\text{after}} = 87.87$, NIS $= 1.410 \leq 2.5$).

On an Intel Core i7 CPU, our online health feature extraction, decoupled eigendecomposition, joint detector, and signature matrix router require $\mathbf{0.0599}$ ms (P50) and $\mathbf{0.0852}$ ms (P95) per window—only $\mathbf{0.0639}$% of a 10 Hz LiDAR frame budget—while adding the 48-cell spherical encoder and 2-layer causal relative-attention network takes $\mathbf{0.633}$–$\mathbf{0.639}$ ms (P50). Delayed-commit queue management and SHA-256 checkpointing take 0.0034 ms (P50) and 0.4171 ms (P95) (48.5 KB per snapshot). In a 10-scan window microbenchmark on the persistent 20 cm global voxel map of `hall_01` ($|M| = 594,979$ voxels, $|\Delta M_{\text{dirty}}| = 82,219$ dirty voxels, 13.8% of the map), full-dict deep-copy snapshot takes 1797.26 ms, whereas our Copy-on-Write `SpatialMutationLog` rolls back the 82,219 dirty voxels in $\mathbf{99.20}$ ms ($\mathbf{18.12}\times$ local rollback speedup) with exact bit-for-bit restoration. The 25k-point RANSAC mixture filter takes 30.82 ms (P50) and is invoked exclusively on quarantined background branches.

VI. LIMITATIONS AND FUTURE WORK

First, as formalized in Proposition 1, when all onboard modalities degrade simultaneously without revisit loops (Scenario R4), no algorithm can synthesize true metric information; our framework guarantees safe containment (Mode 3/4) rather than recovery. Second, while B0, B2, B5, and point-mixture/orthogonal-anchor B6 variants on LIO-SAM, FAST-LIO2, MM-LINS, and ORB-SLAM3 (`ORB_SINGLE_MAP=1`) are executed end-to-end on real host binaries (`REAL_HOST` / `INJECTION`), our keyframe transaction manager currently operates at the keyframe pose-graph and rosbag boundary rather than rolling back internal C++ `ikd-Tree` pointers or GTSAM `ISAM2` Bayes-tree marginals in-place, and pose-graph bridging variants on LIO-SAM and ORB-SLAM3 are evaluated via offline keyframe-graph replay (`OFFLINE_CONSTRUCT`). Embedding the transaction buffer directly inside C++ `ikd-Tree` / `LocalMapping` threads and evaluating the remaining held-out M2DGR test split (`room_01`, `room_dark_02`, `hall_02`, `door_02`, `street_08`) and physical hardware R0–R5 trials are our immediate next steps. Third, because public datasets provide point clouds rather than raw full-waveform LiDAR pulses, our experimental validation operates at the point-mixture level rather than waveform deconvolution.

VII. Conclusion

We presented a keyframe-level transactional recovery framework for degraded multi-sensor SLAM that coordinates online directional reweighting, retrospective quarantine repair, and multi-map hibernation under a four-mode information–uncertainty gate $\mathcal{D}(t)$. By pairing $2\,\mathrm{s}$ delayed-commit buffering with a zero-training mechanism signature router and holdout-screened atomic commit, our approach mitigates pre-hibernation tail drift, overcomes the insensitivity of covariance detectors to near-field self-consistent occlusions, prevents unverified merge degradation, and transfers across heterogeneous LiDAR and visual SLAM hosts.

References

[1] W. Xu, Y. Cai, D. He, J. Lin, and F. Zhang, "FAST-LIO2: Fast direct LiDAR-inertial odometry," *IEEE Transactions on Robotics*, vol. 38, no. 4, pp. 2053–2073, 2022.

[2] T. Shan, B. Englot, D. Meyers, W. Wang, C. Ratti, and D. Rus, "LIO-SAM: Tightly-coupled lidar inertial odometry via smoothing and mapping," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2020, pp. 5135–5142.

[3] C. Campos, R. Elvira, J. J. G. Rodríguez, J. M. M. Montiel, and J. D. Tardós, "ORB-SLAM3: An accurate open-source library for visual, visual–inertial, and multimap SLAM," *IEEE Transactions on Robotics*, vol. 37, no. 6, pp. 1874–1890, 2021.

[4] S. Zhao, H. Zhang, P. Wang, L. Nogueira, and S. Scherer, "Super odometry: IMU-centric LiDAR-visual-inertial estimator for challenging environments," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2021, pp. 8729–8736.

[5] Y. Ma, J. Xu, S. Yuan, T. Zhi, W. Yu, J. Zhou, and L. Xie, "MM-LINS: A multi-map LiDAR-inertial system for over-degenerate environments," *IEEE Transactions on Intelligent Vehicles*, vol. 10, no. 1, pp. 472–482, 2025.

[6] T. Tuna, J. Nubert, Y. Nava, S. Khattak, and M. Hutter, "X-ICP: Localizability-aware LiDAR registration for robust localization in extreme environments," *IEEE Transactions on Robotics*, vol. 40, pp. 452–471, 2024.

[7] J. Zhang, M. Kaess, and S. Singh, "On degeneracy of optimization-based state estimation problems," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2016, pp. 809–816.

[8] T. Shan, B. Englot, C. Ratti, and D. Rus, "LVI-SAM: Tightly-coupled lidar-visual-inertial odometry via smoothing and mapping," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2021, pp. 5692–5698.

[9] G. Kim and A. Kim, "Scan context: Egocentric spatial descriptor for place recognition within 3D point cloud map," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2018, pp. 4802–4809.

[10] H. Huang, W. Lin, and M. Liu, "Dual-SLAM: A framework for robust single-camera SLAM," *IEEE Transactions on Robotics*, vol. 36, no. 4, pp. 1142–1157, 2020.

[11] G. Kim and A. Kim, "LT-mapper: A modular framework for LiDAR-based lifelong mapping," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2022, pp. 7995–8002.

[12] N. Sünderhauf and P. Protzel, "Switchable constraints for robust pose graph SLAM," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2012, pp. 1879–1884.

[13] J. G. Mangelson, D. Dominic, R. M. Eustice, and R. Vasudevan, "Pairwise consistent measurement set maximization for robust multi-robot map merging," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2018, pp. 2916–2923.

[14] H. Yang, P. Antonante, V. Tzoumas, and L. Carlone, "Graduated non-convexity for robust spatial perception: From non-minimal solvers to global outlier rejection," *IEEE Robotics and Automation Letters*, vol. 5, no. 2, pp. 1127–1134, 2020.

[15] J. Yin, A. Li, T. Li, W. Yu, and D. Zou, "M2DGR: A multi-sensor and multi-scenario SLAM dataset for ground robots," *IEEE Robotics and Automation Letters*, vol. 7, no. 2, pp. 2266–2273, 2021.